

American College of Radiology
ACR Appropriateness Criteria®

Right Lower Quadrant Pain

Variant 1: Right lower quadrant pain. Initial imaging.

Procedure	Appropriateness Category	SOE	Adults RRL	Peds RRL	Rating	Median	Final Tabulations								
							1	2	3	4	5	6	7	8	9
CT abdomen and pelvis with IV contrast	Usually appropriate	Strong	⊕⊕⊕ 1-10 mSv	⊕⊕⊕⊕ 3-10 mSv [ped]	9	9	1	0	0	0	0	0	0	5	12
		References		Study Quality											
		29 (26341808)		2											
		33 (25419588)		3											
		27 (23023965)		2											
		26 (19843741)		2											
		34 (19733421)		4											
		31 (2239906)		4											
		28 (31743429)		Good											
		32 (31219315)		4											
		1 (32684379)		3											
		30 (33211574)		3											
MRI abdomen and pelvis without and with IV contrast	May be appropriate	Strong	○ ○ mSv	○ ○ mSv [ped]	6	6	0	0	1	1	4	9	2	1	0
		References		Study Quality											
		40 (26901006)		Good											
		36 (26807893)		2											
		37 (26526902)		3											
		38 (24272981)		2											

		30 (33211574)			3											
Fluoroscopy contrast enema	Usually not appropriate	Expert Consensus	☼☼☼ 1-10 mSv	☼☼☼☼ 3-10 mSv [ped]	2	2	8	5	3	1	0	0	0	0	1	
WBC scan abdomen and pelvis	Usually not appropriate	Limited	☼☼☼☼ 10-30 mSv		2	2	4	6	6	0	1	0	0	0	1	
		References			Study Quality											
		52 (1456603)			3											

Variant 2: Right lower quadrant pain, fever, leukocytosis. Suspected appendicitis. Initial imaging.

Procedure	Appropriateness Category	SOE	Adults RRL	Peds RRL	Rating	Median	Final Tabulations								
							1	2	3	4	5	6	7	8	9
CT abdomen and pelvis with IV contrast	Usually appropriate	Strong	☼☼☼ 1-10 mSv	☼☼☼☼ 3-10 mSv [ped]	8	n/a	0	0	0	0	0	0	0	0	0
		References		Study Quality											
		57 (27318889)		2											
		18 (26518583)		2											
		29 (26341808)		2											
		58 (26248155)		3											
		56 (26031278)		Good											
		62 (25831188)		4											
		66 (25714280)		3											
		8 (25488354)		2											
		59 (25320531)		3											
		67 (25154940)		3											
		61 (25127845)		3											
		64 (24633221)		2											
		55 (24598250)		2											
		53 (23845254)		2											
		54 (22951113)		3											

27 (23023965)	2
63 (22818790)	2
60 (22226440)	4
31 (2239906)	4
10 (21251432)	Good
28 (31743429)	Good
65 (29916776)	Good
30 (33211574)	3

CT abdomen and pelvis without IV contrast	May be appropriate	Strong	☼☼☼ 1-10 mSv	☼☼☼☼ 3-10 mSv [ped]	6	n/a	0	0	0	0	0	0	0	0	0
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References	Study Quality
57 (27318889)	2
18 (26518583)	2
29 (26341808)	2
58 (26248155)	3
56 (26031278)	Good
62 (25831188)	4
66 (25714280)	3
8 (25488354)	2
59 (25320531)	3
67 (25154940)	3
61 (25127845)	3
64 (24633221)	2
55 (24598250)	2
53 (23845254)	2
54 (22951113)	3
27 (23023965)	2
63 (22818790)	2
60 (22226440)	4
31 (2239906)	4
10 (21251432)	Good

		28 (31743429)		Good													
		65 (29916776)		Good													
		30 (33211574)		3													
US abdomen	May be appropriate	Strong	O 0 mSv	O 0 mSv [ped]	6	n/a	0	0	0	0	0	0	0	0	0	0	
		References		Study Quality													
		72 (27203288)		3													
		71 (25771443)		3													
		70 (25475515)		3													
		68 (24768483)		3													
		73 (22638942)		4													
		69 (22360013)		3													
		42 (2934762)		4													
		65 (29916776)		Good													
		74 (31404675)		Good													
MRI abdomen and pelvis without and with IV contrast	May be appropriate	Strong	O 0 mSv	O 0 mSv [ped]	5	n/a	0	0	0	0	0	0	0	0	0	0	
		References		Study Quality													
		40 (26901006)		Good													
		36 (26807893)		2													
		37 (26526902)		3													
		38 (24272981)		2													
		39 (23481162)		2													
		3 (22033948)		2													
		65 (29916776)		Good													
		23 (31420704)		3													
US pelvis	May be appropriate	Limited	O 0 mSv	O 0 mSv [ped]	5	n/a	0	0	0	0	0	0	0	0	0	0	
		References		Study Quality													
		51 (25499245)		3													
		50 (22421290)		3													

		75 (22273324)			3												
MRI abdomen and pelvis without IV contrast	May be appropriate	Strong	O 0 mSv	O 0 mSv [ped]	4	n/a	0	0	0	0	0	0	0	0	0	0	
		References			Study Quality												
		40 (26901006)			Good												
		36 (26807893)			2												
		37 (26526902)			3												
		38 (24272981)			2												
		39 (23481162)			2												
		3 (22033948)			2												
		65 (29916776)			Good												
		23 (31420704)			3												
CT abdomen and pelvis without and with IV contrast	Usually not appropriate	Strong	☢☢☢☢ 10-30 mSv	☢☢☢☢☢ 10-30 mSv [ped]	3	n/a	0	0	0	0	0	0	0	0	0	0	
		References			Study Quality												
		57 (27318889)			2												
		18 (26518583)			2												
		29 (26341808)			2												
		58 (26248155)			3												
		56 (26031278)			Good												
		62 (25831188)			4												
		66 (25714280)			3												
		8 (25488354)			2												
		59 (25320531)			3												
		67 (25154940)			3												
		61 (25127845)			3												
		64 (24633221)			2												
		55 (24598250)			2												
		53 (23845254)			2												
		54 (22951113)			3												

[illegible]

Variant 3: Pregnant woman. Right lower quadrant pain, fever, leukocytosis. Suspected appendicitis. Initial imaging.

Procedure	Appropriateness Category	SOE	Adults RRL	Peds RRL	Rating	Median	Final Tabulations								
							1	2	3	4	5	6	7	8	9
US abdomen	Usually appropriate	Limited	O 0 mSv	O 0 mSv [ped]	8	n/a	0	0	0	0	0	0	0	0	0
		References		Study Quality											
		86 (27554180)		3											
		85 (24151869)		3											
		87 (22370694)		3											
		42 (2934762)		4											

MRI abdomen and pelvis without IV contrast	Usually appropriate	Limited	○ 0 mSv	○ 0 mSv [ped]	7	n/a	0	0	0	0	0	0	0	0	0
		References		Study Quality											
		82 (26338256)		3											
		80 (26215327)		3											
		81 (25291255)		3											
		78 (25148766)		3											
		83 (23360736)		3											
		84 (-3188535)		4											
		76 (29869102)		3											
US pelvis	May be appropriate	Expert Consensus	○ 0 mSv	○ 0 mSv [ped]	6	n/a	0	0	0	0	0	0	0	0	0
CT abdomen and pelvis with IV contrast	May be appropriate	Limited	⊗⊗⊗ 1-10 mSv	⊗⊗⊗⊗ 3-10 mSv [ped]	5	n/a	0	0	0	0	0	0	0	0	0
		References		Study Quality											
		78 (25148766)		3											
		33 (25419588)		3											
		79 (29948087)		4											
CT abdomen and pelvis without IV contrast	May be appropriate	Limited	⊗⊗⊗ 1-10 mSv	⊗⊗⊗⊗ 3-10 mSv [ped]	4	n/a	0	0	0	0	0	0	0	0	0
		References		Study Quality											
		78 (25148766)		3											
		33 (25419588)		3											
		79 (29948087)		4											
CT abdomen and pelvis without and with IV contrast	Usually not appropriate	Limited	⊗⊗⊗⊗ 10-30 mSv	⊗⊗⊗⊗⊗ 10-30 mSv [ped]	3	n/a	0	0	0	0	0	0	0	0	0
		References		Study Quality											
		78 (25148766)		3											
		33 (25419588)		3											

Appendix Key

A more complete discussion of the items presented below can be found by accessing the supporting documents at the designated hyperlinks.

Appropriateness Category: The panel's recommendation for a procedure based on the assessment of the risks and benefits of performing the procedure for the specified clinical scenario.

SOE: Strength of Evidence. The assessment of the amount and quality of evidence found in the peer reviewed medical literature for an appropriateness recommendation.

- **References:** The citation number and PMID for the reference(s) associated with the recommendation.
- **Study Quality:** The assessment of the quality of an individual reference based on the number of study quality elements described in the reference.

RRL: Relative Radiation Level. A population based assessment of the amount of radiation a typical patient may be exposed to during the specified procedure.

Rating: The final rating (1-9 scale) for the procedure as determined by the panel during rating rounds.

Median: The median rating (1-9 scale) for the procedure as determined by the panel during rating rounds.

Final tabulations: A histogram showing the number of panel members who rated the procedure as noted in the column heading (ie, 1, 2, 3, etc.).

Additional supporting documents about the AC methodology and processes can be found at www.acr.org/ac.